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— MEDAN ELEKTROMAGNETIK 2 —

— Gradien —

diberlakukan saat besaran skalar hasil vektor

↳ memiliki arah + Magnetudo

↓

posisi max

dari potensial listrik

→ klo gradien thdp potensial listrik = 0 → maka potensial listrik konstan

→ $\vec{E}$, $\vec{H}$, $\vec{D}$, $\vec{B}$

— Divergensi —

diberlakukan pada besaran vektor hasil skalar

→ mengetahui disuatu tempat ada sumber / tidak

sumber 2 < listrik / magnetik → arus listrik → bergerak → menurunkan arah

Divergensi thdp kerapatan fluks magnetik = 0

$\nabla \cdot \vec{D}$ → do dot del vektor

Sistem koordinat kartesian → $\nabla \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$ → klo thdp x berarti ke x aja, y ke y aja, z ke z aja

Sistem koordinat silinder → $\nabla \cdot \vec{D} = \frac{1}{\rho} \frac{\partial (\rho D_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$

Sistem koordinat bola → $\nabla \cdot \vec{D} = \frac{1}{r^2} \frac{\partial (r^2 D_r)}{\partial r} + \frac{1}{r \sin\theta} \frac{\partial (\sin\theta D_\theta)}{\partial \theta} + \frac{1}{r \sin\theta} \frac{\partial D_\phi}{\partial \phi}$

contoh kartesian :

① $\vec{D} = 2xy^2 \, \tilde{a}x$ → $D_x = 2xy^2$

$D_y = 0$

$D_z = 0$

$\nabla \cdot \vec{D} = \frac{\partial (2xy^2)}{\partial x} + \frac{\partial (0)}{\partial y} + \frac{\partial (0)}{\partial z}$

$= 2y^2$,

$\nabla \cdot \vec{D}$ misal di titik observator P (x = 2 ; y = 1, z = 3)

maka $\nabla \cdot \vec{D} = 2(1)^2$

$= 2$ // → klo ≠0 maka dititik observartor ada sumber listrik

Dipindai dengan CamScanner

② $\vec{D} = 2y^2 z\, \hat{a}_x + 3xy^3 z\, \hat{a}_z$

$D_x = 2y^2 z\, \hat{a}_x$

$D_y = 0$

$D_z = 3xy^3 z\, \hat{a}_z$

$\nabla . \vec{D} = \frac{\partial (2y^2 z)}{\partial x} + \frac{\partial (0)}{\partial y} + \frac{\partial (3xy^3 z)}{\partial z}$

$= 0 + 0 + 3xy^3$

$= 3xy^3$

Contoh silinder

$\vec{D}_{silinder} = D_\rho \hat{a}_\rho + D_\phi \hat{a}_\phi + D_z \hat{a}_z$ (coloumb/m^2)

① $\vec{D} = 2\sin\phi\, \hat{a}_\phi$

$D_\rho = 0$

$D_\phi = 2\sin\phi$

$D_z = 0$

$\nabla . \vec{D} = \frac{1}{\rho} \frac{\partial (0)}{\partial \rho} + \frac{1}{\rho} \frac{\partial (2\sin\phi)}{\partial \phi} + \frac{\partial 0}{\partial z}$

$= 0 + 2\cos\phi + 0$

$= 2\cos\phi$

② $\vec{D} = \rho^2 \sin\phi\, \hat{a}_\rho + 3z^2 \hat{a}_\phi + 3\sin\phi\, \hat{a}_z$ (c/m^2)

$\nabla . \vec{D}$ di titik A ($x=2, y=1, z=0$)

$D_\rho = \rho^2 \sin\phi$

$D_\phi = 3z^2$

$D_z = 3\sin\phi$

$\nabla . \vec{D} = \frac{1}{\rho} \frac{\partial (\rho D_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$

$= \frac{1}{\rho} \frac{\partial (\rho^3 \sin\phi)}{\partial \rho} + \frac{1}{\rho} \frac{\partial (3z^2)}{\partial \phi} + \frac{\partial (3\sin\phi)}{\partial z}$

$= \frac{1}{\rho} (3\rho^2 \sin\phi) + 0 + 0$

$= \frac{1}{\rho} 3\rho^2 \sin\phi$

$= 3\rho \sin\phi$

Dipindai dengan CamScanner

Hitung kartesian A (x=2, y=1, z=0) → ubah ke silinder

$\rho = \sqrt{x^2+y^2} = \sqrt{5}$

$\phi = \tan^{-1}\left(\frac{y}{x}\right) = 26{,}57°$

$z = 0$

$\nabla \cdot \vec{D} = 3\rho \sin\phi$

$= 3.\sqrt{5} \sin 26{,}57°$

$= 6{,}71 \sin 26{,}57$

$= 3{,}001$

↳ krn (+) berarti ada sumber kerapatan fluks listrik

contoh bola

$\vec{D}$ bola $= D_r \hat{a}_r + D_\theta \hat{a}_\theta + D_\phi \hat{a}_\phi$ observator $P(r,\theta,\phi)$

① $\vec{D} = 2\sin\theta \sin\phi \, \hat{a}_\theta$

$\nabla \cdot \vec{D}$?

$D_r = 0$

$D_\theta = 2 \sin\theta \sin\phi$

$D_\phi = 0$

$$\nabla \cdot \vec{D} = \frac{1}{r^2}\frac{\partial (r^2 D_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial (\sin\theta D_\theta)}{\partial \theta} + \frac{1}{r\sin\theta}\frac{\partial D_\phi}{\partial \phi}$$

$$= \frac{1}{r^2}\frac{\partial (0)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial (2\sin^2\theta \sin\phi)}{\partial \theta} + \frac{1}{r\sin\theta}\frac{\partial (0)}{\partial \phi}$$

$$= 0 + \frac{2\sin\phi}{r\sin\theta}\frac{\partial (\sin^2\theta)}{\partial \theta} + 0$$

$$= \frac{2\sin 2\theta \sin\phi}{r\sin\theta}$$

$$= \frac{2}{r}\frac{(2\sin\theta\cos\theta)\sin\phi}{\sin\theta}$$

$$= \frac{4\cos\theta\sin\phi}{r}$$

Dipindai dengan CamScanner

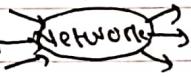
Divergensi → u/ melihat ada sumber / tidak

Fluks in 

Fluks out $\nabla \cdot \vec{D} > 0$ → berarti ada sumber listrik di daerah tsb

Fluks in 

Fluks out $\nabla \cdot \vec{D} < 0$ → ada absorber (penyerap elektron)
yang membuat fluks listrik yg keluar menjadi lebih kecil dari fluks masuk

$\nabla \cdot \vec{D} = 0$ fluks in 

fluks out → masuk = keluar tidak ada sumber apapun, tidak ada absorber, tidak ada apa-apa

LATIHAN SOAL

1. $\vec{D} = \tilde{a}_y$

$D_x = 0$

$D_y = 1$

$D_z = 0$

$$\nabla \cdot \vec{D} = \frac{\partial(0)}{\partial x} + \frac{\partial(1)}{\partial y} + \frac{\partial(0)}{\partial z}$$

$$= 0 + 0 + 0$$

$$= 0 \text{ //}$$

2. $\vec{D} = x\,\tilde{a}_y$

$D_x = 0$

$D_y = x$

$D_z = 0$

$$\nabla \cdot \vec{D} = \frac{\partial(0)}{\partial x} + \frac{\partial(x)}{\partial y} + \frac{\partial(0)}{\partial z}$$

$$= 0 + 0 + 0$$

$$= 0 \text{ //}$$

Dipindai dengan CamScanner

3. $\vec{B} = x\vec{a}_x$

$\nabla \cdot \vec{B} = 0$ → Dikarenakan Divergensi terhadap kerapatan Fluks magnetik = 0

4. $\vec{D} = \frac{1}{\rho}\vec{a}_\rho$

$D_\rho = \frac{1}{\rho}$

$D_\phi = 0$

$D_z = 0$

$\nabla \cdot \vec{D} = \frac{1}{\rho}\frac{\partial(\rho \cdot \frac{1}{\rho})}{\partial \rho} + \frac{1}{\rho}\frac{\partial(0)}{\partial \phi} + \frac{\partial(0)}{\partial z}$

$= \frac{1}{\rho} \cdot 0 + 0 + 0$

$= 0$ //

5. $\vec{B} = \vec{a}_r$

$\nabla \cdot \vec{B} = 0$ → Dikarenakan Divergensi terhadap kerapatan Fluks magnetik = 0

6. $\vec{D} = 3\rho\vec{a}_\rho + 6\vec{a}_z$

$D_\rho = 3\rho$

$D_\phi = 0$

$D_z = 6$

$\nabla \cdot \vec{D} = \frac{1}{\rho}\frac{\partial(\rho \cdot 3\rho)}{\partial \rho} + \frac{1}{\rho}\frac{\partial(0)}{\partial \phi} + \frac{\partial(6)}{\partial z}$

$= \frac{1}{\rho}\frac{\partial(3\rho^2)}{\partial \rho} + \frac{1}{\rho}\frac{\partial(0)}{\partial \phi} + \frac{\partial(6)}{\partial z}$

$= \frac{1}{\rho}(6\rho) + 0 + 0$

$= 6$ //

Dipindai dengan CamScanner